Project: Worked examples
Title: PCB trace width (IPC-2221 legacy)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The minimum width of a printed-circuit trace for a current and an allowed temperature rise, from the legacy IPC-2221 curve-fit, and the resistance, voltage drop and I^2R loss of the trace as drawn: legacy IPC-2221 curve-fit, conservative for multilayer boards. External or internal layer; copper weight in oz/ft²; resistance at the trace's operating temperature (ambient plus rise). The IPC-2152 method (charts, plane and board-thickness factors) is not implemented.

WHAT IT CALCULATES	RELATION
Trace cross-section	I = k·ΔT^0.44·A^0.725, so A = [I/(k·ΔT^0.44)]^(1/0.725) with A in mil², k = 0.048 external, 0.024 internal
Copper thickness	$t = w_{Cu} \cdot t_{oz}$, t_{oz} = 1.378 mil per oz/ft² (35 µm, derived)
Minimum width	$w_{min} = A/t$; utilisation $u_w = w_{min}/w_{act}$
Resistivity at the operating temperature	$\rho_T = \rho_{20} \cdot [1 + \alpha \cdot (T_{amb} + \Delta T - 20)]$
Trace resistance	$R = \rho_T \cdot L/(w_{act} \cdot t)$; voltage drop ΔV = I·R; loss P = I²·R

Basis: Electrical properties: current-carrying capacity of printed-board conductors (legacy curve-fit)[1](6); Resistivity and temperature coefficient of resistance[2](Ch. on DC circuits).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I	Trace current	5	A	spec: Driver datasheet DD-12: 5 A continuous motor supply	0.01 … 35 A
dT	Allowed temperature rise above ambient	20	K	spec: Board design rule BDR-3: 20 K allowed rise on power rails	1 … 100 K
T_{amb}	Ambient temperature at the board	25	°C	spec: Environmental specification ES-2: 25 °C at the board	-40 … 125 °C
layer	Layer	External layer		drawing: PCB-LAY-014 rev B	any
Cu_{oz}	Finished copper weight in oz/ft² (0.5 to 6), not a thickness in µm: 17.5 µm is ½ oz, 35 µm is 1 oz, 70 µm is 2 oz, 105 µm is 3 oz	2		spec: Fabrication note FN-7: 2 oz/ft² finished copper on the outer layers	0.5 … 6
w_{act}	Trace width as drawn	2.5	mm	drawing: PCB-LAY-014 rev B	0.05 … 10 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Trace length	100	mm	drawing: PCB-LAY-014 rev B	1 … 1000 mm
k_{ext}	Curve-fit constant k, external layer	0.048		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.01 … 0.1
k_{int}	Curve-fit constant k, internal layer	0.024		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.005 … 0.05
n_{dT}	Exponent of the temperature rise	0.44		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.3 … 0.6
n_A	Exponent of the cross-section	0.725		reference: record: IPC-2221 legacy curve-fit constants (published closed form; the issuer enters the edition)	0.5 … 1
t_{oz}	Copper thickness per oz/ft² (35 µm = 1.378 mil, derived)	1.378	mil	reference: record: IPC-2221 legacy curve-fit constants; 1.378 mil = 35 µm per oz/ft², derived	1.2 … 1.5 mil
ρ_{20}	Copper resistivity at 20 °C (annealed copper, 1.724e-8 Ω·m)	0.01724	ohm·mm²/ft	reference: record: annealed copper 1.724e-8 Ω·m at 20 °C, derived physical constant	0.0165 … 0.0185 ohm·mm²/m
α	Temperature coefficient of resistance of copper	0.00393	1/K	reference: record: annealed copper, 0.00393 per K at 20 °C, derived physical constant	0.0035 … 0.0042 1/K

3. Assumptions

A1. The curve-fit is the legacy IPC-2221 relation, drawn for a trace on an outer or inner layer with no plane or convective help, 1 to 3 oz copper, up to 35 A, rises of 10 to 100 K and widths to 10 mm; it is conservative for multilayer boards and is a screening of width, not a thermal analysis of the board. (IPC-2221 curve-fit (legacy); IPC-2152 is not implemented.)

A2. Copper thickness is the nominal 1.378 mil (35 µm) per oz/ft² of finished copper; plating build-up on outer layers and etch-back are not modelled. (Derived: 35 µm / 25.4 µm per mil.)
A3. Resistance uses the trace as drawn at the temperature ambient + rise, from the resistivity at 20 °C and a linear temperature coefficient; vias, pads, bends and skin effect are ignored. (Derived constants for annealed copper.)
A4. Steady direct current (or RMS current at low frequency); the rise is the allowed rise in the worst case. (Scope of the curve-fit.)

4. Working

[14] $k = k_{ext}$ [1](§6 (legacy))
= 0.048
= 0.048
Curve-fit constant k (external layer)

[15] $I_n = I / (1\ A)$
= 5 A / 1 A
= 5
Current as a number of amperes

[16] $dT_n = dT / (1\ K)$
= 20 K / 1 K
= 20
Temperature rise as a number of kelvin

[17] $A_{req} = (I_n / (k * dT_n^{n_{dT}}))^{(1 / n_A)} * (1\ mil^2)$ [1](§6 (legacy))
= (5 / (0.048 × 20^0.44))^(1 / 0.725) × 1 mil^2
= 98.511 mil²
Minimum conductor cross-section

[18] $t_{Cu} = Cu_{oz} * t_{oz}$
= 2 × 1.378 mil
= 70.002 um
Copper thickness

[19] $w_{min} = A_{req} / t_{Cu}$
= 98.511 mil² / 70.002 um
= 0.9079 mm
Minimum trace width

[20] $u_w = w_{min} / w_{act}$
= 0.9079 mm / 2.5 mm
= 0.36316
Width utilisation (minimum / as drawn)

[21] $T_{trace} = T_{amb} + dT$
= 25 degC + 20 K
= 45 degC
Trace operating temperature

[22] $\rho_T = \rho_{20} * (1 + \alpha * (T_{trace} - 20\ degC))$
= 0.01724 ohm·mm²/m × (1 + 0.00393 1/K × (45 degC - 20 degC))
= 0.018934 ohm·mm²/m
Copper resistivity at the operating temperature

[23] $A_{act} = w_{act} * t_{Cu}$
= 2.5 mm × 70.002 um
= 0.17501 mm²
Cross-section of the trace as drawn

[24] $trace_R = \rho_T * L / A_{act}$
= 0.018934 ohm·mm²/m × 100 mm / 0.17501 mm²
= 10.819 mohm
Trace resistance at the operating temperature

[25] vdrop = I * $trace_R$
= 5 A × 10.819 mohm
= 54.095 mV
Voltage drop along the trace
[26] $trace_{Ploss}$ = I^2 * $trace_R$
= (5 A)^2 × 10.819 mohm
= 270.47 mW
Power loss in the trace (I^2R)

5. Results

SYMBOL	VALUE	DESCRIPTION
A_{req}	98.511 mil²	Minimum cross-section
w_{min}	0.9079 mm	Minimum trace width
u_w	0.36316	Width utilisation
$trace_R$	10.819 mohm	Trace resistance
vdrop	54.095 mV	Voltage drop
$trace_{Ploss}$	270.47 mW	Power loss

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Trace width as drawn against the minimum for the rise	u_w = 0.36316	$u_w \le 1$	0.363	[1](§6 (legacy))

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: k_{ext}, k_{int}, n_{dT}, n_A, t_{oz} (IPC-2221 legacy curve-fit constants); ρ_{20}, α (Copper resistivity constants). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template elec.pcb-trace@1.0.3 Content hash 0d7d5417226f1fcc Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. IPC-2221.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.